

LangGap: Diagnosing and Closing the Language Gap in Vision-Language-Action Models

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github.com/YC11Hou/langgap



Code & benchmark

Diagnosis
99 Tasks · 4 Dimensions · 1,980 Episodes

Semantic Perturbation

Same Scene · Different Words · Language-Only Signal

$\pi 0.5$: 93.8% on the original instruction → 21.4% after one changed word

Change Object 29.3% · Change Target 0.0% · Spatial Description 11.0% · Drawer Action 31.7%



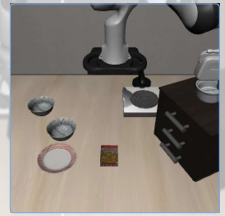
Training
16 Tasks · 2,400 Demos · LoRA

Scripted Demonstrations

Waypoint Controllers · Verified · Reproducible

0% → 90% single-task; gain does not scale

OSC_POSE · verified · 16/43 split



Open Benchmark

github.com/YC11Hou/langgap

- 99 tasks · 59 semantic variants
- Instruction-level split 16 / 43
- Physically validated in LIBERO
- 2,400 demos + LoRA recipes
- 4 VLA models evaluated



Diagnose Benchmark Train No Shortcuts

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The Language Gap

— 72.4 pts

93.8% → 21.4% · Change Target 0% on all 13 tasks

Orig. 93.8

Obj. 29.3

Tgt. 0.0

Spat. 11.0

Drawer 31.7

Instruction VLA ($\pi 0.5$) Success?

Leaderboard

59 Ext. Tasks · 20 Ep./Task

	Ch. Obj.	Ch. Tgt.	Sp. Desc.	Drawer	Total
$\pi 0.5$	29.3	0.0	11.0	31.7	21.4
$\pi 0$	10.8	0.0	20.0	0.0	8.6
$\pi 0$ -FAST	3.1	2.3	2.5	0.0	2.7
SmoTVLA	7.6	0.0	8.0	16.7	6.4
Ours (45-task)	28.4	6.2	25.0	20.0	22.8
Ours (56-task)	27.5	5.0	17.0	3.3	20.4

Benchmark Highlights

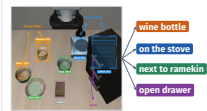
4 LIBERO Suites · 99 Tasks · Scripted Data



Every alternative object, target and drawer named by an extended instruction is physically present; each extended task keeps the exact initial state of its source task.

1 How LangGap is built: three design principles

1 Same-scene multi-task



one scene, h instructions

h tasks share one identical initial state, so a language-blind model is capped at 1/ h : no visual confounds.

2 Instruction-level split



59 extended tasks · 16 train · 43 held-out

Train on 16 (9 spatial · 4 object · 3 goal); test on 43 held-out tasks whose instructions are **never seen in training**.

3 Feasibility validation

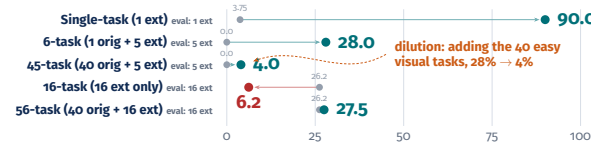
- graspable object can be picked
- reachable target within reach
- detectable success observable

Every candidate runs in the LIBERO simulator; only the 59 passing all three checks enter the benchmark.

LIBERO: 130 tasks, 0 semantic · LIBERO-Plus: 10,030 tasks, 1 of 7 dims linguistic · LangGap: 99 tasks, 59 semantic, compositional

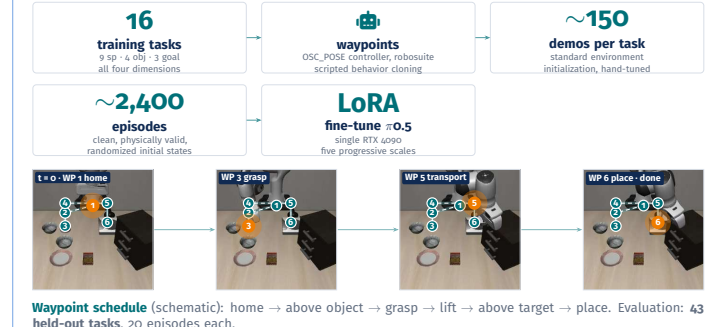
2 Can data close the gap? The dilution effect

LoRA on $\pi 0.5$, five scales, extended tasks: ● base → ● tuned (● worse)



- Dilution: +40 official tasks drops 28% → 4% on the same eval — easy visual tasks dilute the language-grounding signal.
- Capacity wall: 90% (1 task) → 28% (6) → 6.2% (16) — diverse semantics is the fundamental challenge.
- Single-task fine-tuning on one scripted task lifts an unseen-instruction success rate from 0% to 90% (frames at left).

3 Training-data collection at scale



Failure is structured: the spatial-goal slot collapses to 0%; object nouns only partially ground.

Data narrows the gap (0% → 90% single-task) — but does not scale to diverse semantics.

Measure language generalization before deployment — LangGap is the yardstick, released with code + data.